Subject Description Form

Subject Code	ITC6982RT
Subject Title	Guided Study in Biomimicry in Fashion
Credit Value	3
Level	6
Pre-requisite / Co-requisite / Exclusion	Nil
Objectives	Guide students to carry out interdisciplinary and cost-effective researches, and to resolve problems with adaptable and sustainable approaches. z
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. comprehend biomimetic fashion from a view of adaptable and sustainable system; b. implement an interdisciplinary research in fashion field through a holistic understanding of nature ecosystem; c. consciously and actively align their researches on biomimetic fashion to a natural evolution way; d. creatively design sophisticated green approaches to solve human problems in fashion design and manufacturing field; a. continuously develop their knowledge and ability of harmonizing artificial systems with living nature environment via biomimicry.
Subject Synopsis/ Indicative Syllabus*)*	a. Principle of biomimicry: adaptable & sustainable approach. b. Biomimetic fashion with diverse fibers in nature: fiber spinning & extraction. c. Biomimetic fashion with functional surfaces in nature: surface energy & smoothness. d. Biomimetic fashion with natural thermal insulation: porosity and orientation. e. Biomimetic fashion with natural optical systems: refractive index and band gap. f. Complexity in fashion biomimicry: artificial vs. living.
Teaching/Learning Methodology	Combination of group discussion and self-learning, reference investigation and hand practice, case study and presentation, inspiration and reflection via literature review and writing papers.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d	e	
1. Continuous assessment	40%	✓	✓	✓	✓	✓	
2. Open-book examination	60%	✓	✓	✓	✓	✓	
Total	100%						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Through personal assignments & presentations, and individual final exam, all the above learning outcomes can be assessed based on students' behaviors.

Student Study Effort Expected

Class contact:	
▪ Tutorials/Group discussion	26 Hrs.
Other student study effort:	
▪ assignments & presentations	44 Hrs.
▪ reference investigation	50 Hrs.
Total student study effort	120 Hrs.

Reading List and References

1. Bhushan, B. 2009 Biomimetics: lessons from nature—an overview. Phil. Trans. R. Soc. A 367, 1445–1486.
2. Dawson, C., Vincent, J. F. V. & Rocca, A. 1997 How pine cones open. Nature 390, 668.
3. Koch, K., Bhushan, B. & Barthlott, W. 2009 Multifunctional surface structures of plants: an inspiration for biomimetics. Prog. Mater. Sci. 54, 137–178.
4. Vukusic, P. & Sambles, J. R. 2003 Photonic structures in biology. Nature 424, 852–855.
5. Jane Wood. 2019 Bioinspiration in Fashion -- A Review. Biomimetics 4, 16.
6. https://motif.org/news/biomimicry-fashion/
7. https://www.dezeen.com/2021/11/15/auroboros-biomimicry-dress-crystalises-transforms-real-time/